

Distributed Formation and Reconfiguration Control of VTOL UAVs

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Abstract—In this brief, a novel distributed cascade robust feedback control approach is proposed for formation and reconfiguration control of a team of vertical takeoff and landing (VTOL) unmanned air vehicles (UAVs). This approach is based on dynamic communication network. It guarantees intervehicle collision avoidance and considers dynamic constraints of UAVs. In the outer loop of the cascade formation control, a potential field approach is used to generate a desired velocity for each UAV, which ensures that the team of UAVs can perform formation flying, formation rotating and reconfiguration, avoid intervehicle collision, as well as track a specified virtual leader. In the inner loop of the cascade formation control, the velocity of each UAV is designed to track its desired velocity generated by the outer loop, subject to dynamic constraints. The proposed approach is demonstrated via both simulation and flight test.

Index Terms—Distributed control, formation control, multi-agent system, networked control, unmanned air vehicles (UAV).

I. INTRODUCTION

FORMATION control is an important research area of cooperative control of multiple unmanned aerial vehicles (UAVs). The main goal is to coordinate a group of UAVs to achieve a desired formation pattern and avoid intervehicle collisions at the same time. A team of low-cost UAVs flying in formation can replace an expensive multifunction UAV to enable large-scale distributed sensing with high precision, reduce total mission costs, and increase mission success rates. By providing system redundancy, reconfiguration ability, and structure flexibility, formation flight can increase system robustness, reliability, and efficiency. Formation flight of a team of UAVs has broad applications such as terrain and utilities inspection, search and rescue, disaster monitoring, aerial mapping, traffic monitoring, reconnaissance mission, and surveillance.

Motivated by the wide range of possible military and civilian applications, formation control of UAVs has attracted a considerable amount of research interest in the past decade. According to different criteria, formation control approaches in the existing literature have different classifications.

From the control structure perspective, the existing formation control approaches can be classified into the cen-

tralized method, where a single controller is used to control the whole team based on the information from the whole team [13] and the distributed/decentralized method, where each team member generates its own control based on local information from its neighbors [1], [14], [18], [24], [25], [30]. The centralized formation control can be a good strategy for a small team of UAVs. When considering a team with a large number of UAVs, the need for greater computational capacity and a large communication bandwidth would mandate a distributed/decentralized formation control. Over the past ten years, significant efforts have been devoted to dealing with the distributed control problems caused by information constraints. The adaptive gradient climbing strategy [22], linear algebra and graph theory [7], [12], [15], [17], [25], passivity and dissipativity theory [2], Lyapunov methods [11], [28]–[30], and nonlinear small-gain theorem [8] have been introduced to solve the distributed/decentralized formation control problems [18].

From the measurement information perspective, there are two dominant formation control problem formulations [21]: 1) displacement-based and 2) distance-based methods. In displacement-based methods [12], [17], [25], UAVs measure the relative positions of their neighbors based on a common sense of orientation (e.g., global coordinate system). It is well known that graph theory-based consensus control using relative positions ensures global asymptotic convergence to the desired formation. In distance-based methods [5], UAVs measure the relative positions of their neighbors only based on their local reference frames. As UAVs control their formation using distances, formation control becomes complicated.

From the control mechanism perspective, formation control approaches can be classified into consensus-based approaches [7], [12], [15], [17], [25], artificial potential function-based approaches [6], [14], leader–follower approaches [4], [9], [10], [18], [19], virtual structure approaches [24], [26], and behavior-based approaches [3], [20]. Consensus-based approaches convert the formation control problem into the consensus (or stability) problem of relative positions and velocities of multiagents. They achieve formation stability based on graph theory and consensus. However, intervehicle collisions are not considered. Artificial potential function-based approaches apply the negative gradient of a mixture of attractive and repulsive potential functions as control inputs to satisfy the convergence and noncollision properties, respectively. The main drawback of this type of approaches is the appearance of equilibria, where the composite vector field vanishes and the UAVs can get trapped at undesired equilibrium points. Leader–follower approaches simplify the formation problem into individual tracking problems. The main disadvantage is that the leader is a single point of failure for the formation. Moreover, it is difficult for leader–follower approaches to realize

Manuscript received September 17, 2015; accepted February 27, 2016. Date of publication April 14, 2016; date of current version December 14, 2016. Manuscript received in final form March 16, 2016. Recommended by Associate Editor C. De Persis.

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This paper has supplementary downloadable material available at <http://ieeexplore.ieee.org>, provided by the author. The material includes two videos. The total size of the videos is 152 MB. Contact tsllf@nus.edu.sg for further questions about this work.

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Digital Object Identifier 10.1109/TCST.2016.2547952

formation reconfiguration. In virtual structure approaches, the entire formation is considered as a rigid body. The control law of each agent is derived by the following three steps:

- 1) defining the dynamics of the virtual structure;
- 2) translating the motion of the virtual structure into the desired motion of each UAV;
- 3) designing the control law of each UAV to track its desired motion.

As they are centralized and open-loop control approaches, the computation and communication load of the central control is heavy and UAVs may get out of formation if the virtual structure moves too fast to track or the system is affected by disturbance. The behavior-based approach combines the outputs of multiple controllers designed for achieving different and possibly competing behaviors, such as collision avoidance, goal seeking, trajectory tracking, and formation keeping behaviors through suitable weight coefficients that set the relative priority among them. It is natural to adopt this strategy when UAVs have multiple objectives. However, the behaviors may destructively interfere with each other and it is difficult in mathematical analysis.

As mentioned above, the existing approaches have more or less drawbacks in complying with the three fundamental requirements of formation control of UAVs: 1) global convergence to a desired formation; 2) intervehicle collision avoidance; and 3) path following/trajectory tracking. Moreover, from a practical point of view, most of the existing approaches do not consider measurement errors and dynamic constraints of UAVs.

In this brief, we propose a new unified distributed robust control approach for formation and reconfiguration control of UAVs. This approach is based on dynamic undirected communication network, and falls into potential function-based approaches. An identical cascade control consisting of an outer loop and an inner loop is adopted for each individual UAV. The outer loop is based on local relative position measurements (displacement based) and artificial potential functions (potential function based). The artificial potential functions are as follows.

- 1) A graph-based potential function is introduced to achieve formation and reconfiguration control of UAVs;
- 2) an exponential form of potential function is applied to avoid intervehicle collision;
- 3) and a tracking error-based potential function is used to track the desired path/trajectory.

For each UAV, the inner loop is designed to track the desired velocity generated by the outer loop robustly against measurement errors and dynamic constraints. Unlike velocity tracking control of a single UAV, a consensus-based velocity tracking control approach is proposed for a team of UAVs to first achieve consensus on the velocity errors between the actual velocity and the desired velocity of each UAV, and then to drive such consensus value on the velocity errors to zero. The designed controller is based on an analytical solution and ensures that a team of UAVs asymptotically converges to a desired formation and follows a given path/trajectory without intervehicle collisions.

The rest of this brief is organized as follows. Section II presents the formation and reconfiguration controller design. The simulation results of formation flight are shown in Section III, and the experimental results are presented in Section IV, followed by a brief conclusion in Section V.

II. DISTRIBUTED FORMATION AND RECONFIGURATION CONTROL

In this section, a unified distributed robust formation and reconfiguration control approach is proposed to comply with the three fundamental requirements of formation control: 1) global convergence to a desired formation; 2) intervehicle collision avoidance; and 3) path following/trajectory tracking. Both measurement errors and dynamic constraints are considered.

Denote $\mathcal{G} \triangleq (\mathcal{V}, \mathcal{E})$ the dynamic undirected communication network by the vertex set $\mathcal{V} \triangleq \{1, 2, \dots, n\}$, representing the UAVs and the edge set $\mathcal{E} \triangleq \{e_{ij} : i = 1, 2, \dots, n, j \in \mathcal{N}_i\}$ representing the communication links among the UAVs, where the dynamic neighbor set $\mathcal{N}_i$ of UAV i is defined by

$$\mathcal{N}_i \triangleq \{j \in \mathcal{V} : \text{UAV } i \text{ receives information from UAV } j, i \neq j\}. \quad (1)$$

The graph Laplacian $\mathbf{L}_{\mathcal{G}}(t)$ of the dynamic network $\mathcal{G}$ is

$$[\mathbf{L}_{\mathcal{G}}(t)]_{ij} \triangleq \begin{cases} -w_{ij}(t), & \text{if } i \neq j \\ \sum_{k=1, k \neq i}^n w_{ik}(t), & \text{if } i = j \end{cases} \quad (2)$$

with

$$w_{ij}(t) = \begin{cases} 1, & \text{if } j \in \mathcal{N}_i \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where a weight $w_{ij}(t)$ is assigned to each communication link e_{ij} . To ensure global convergence to a desired formation, we assume the following.

Assumption 1 [23]: The dynamic undirected communication network $\mathcal{G}$ is connected, namely, $\text{rank}(\mathbf{L}_{\mathcal{G}}(t)) = n - 1$.

To avoid collision among UAVs, we define the collision avoidance set of UAV i as

$$\mathcal{D}_i \triangleq \{j \in \mathcal{V} : d_{ij}(t) \leq R_a, i \neq j\} \quad (4)$$

where

$$d_{ij}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| \quad (5)$$

with $\mathbf{p}_i(t)$ and $\mathbf{p}_j(t) \in \mathbb{R}^3$ as the positions of UAVs i and j , respectively, and R_a is the maximum collision detection distance.

In addition, a virtual leader (VL) that flies on a desired path/trajectory is introduced in this brief to lead the UAVs in formation to fly along the desired path/trajectory.

The holonomic kinematic models of a team of VTOL UAVs are considered as

$$\begin{cases} \dot{\mathbf{p}}_i(t) = \mathbf{v}_i(t) \\ \dot{\mathbf{v}}_i(t) = \mathbf{u}_i(t) \end{cases} \quad i = 1, 2, \dots, n \quad (6)$$

with $\mathbf{p}_i(t)$ and $\mathbf{v}_i(t) \in \mathbb{R}^3$ the position and velocity of UAV i in north-east-down (NED) frame and $\mathbf{u}_i(t) \in \mathbb{R}^3$ the control.

A. Potential Function-Based Outer-Loop Control

Consider the potential function $V(t)$ for a team of n UAVs

$$V(t) = V^f(t) + V^a(t) + V^t(t) \quad (7)$$

where $V^f(t)$, $V^a(t)$, and $V^t(t) \in \Re$ are the potential functions of formation/reconfiguration, intervehicle collision avoidance, and path/trajjectory tracking of the VL, respectively.

The potential function of formation $V^f(t)$ is defined by

$$V^f(t) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1, j \neq i}^n w_{ij}(t) V_{ij}^f(t) \quad (8)$$

where the weight w_{ij} is defined as in (3) and $V_{ij}^f(t) \in \Re$ is the formation potential function of vehicles i and j , given by

$$V_{ij}^f(t) = \frac{1}{2} [\mathbf{p}_i(t) - \mathbf{p}_j(t) - (\mathbf{p}_{fi} - \mathbf{p}_{fj})]^T \times [\mathbf{p}_i(t) - \mathbf{p}_j(t) - (\mathbf{p}_{fi} - \mathbf{p}_{fj})] \quad (9)$$

with $\mathbf{p}_i(t)$ and $\mathbf{p}_j(t) \in \Re^3$ the positions of UAVs i and j in the NED frame and $\mathbf{p}_{fi}$ and $\mathbf{p}_{fj} \in \Re^3$ the assigned positions of UAVs i and j in the desired formation configuration, which is a step function of time (see Remark 3). It is emphasized that the desired formation configuration $\mathbf{p}_f = [\mathbf{p}_{f1} \ \mathbf{p}_{f2} \ \cdots \ \mathbf{p}_{fn}]^T$ will be maintained if the potential function $V^f(t) = 0$ and the dynamic communication network $\mathcal{G}$ is connected [23]. As is well known, consensus on the relative positions of the UAVs ensures formation keeping. The potential function $V^f(t)$ can be considered as a description of the consensus-based approach in the form of the potential function.

The potential function of intervehicle collision avoidance $V^a(t)$ in (7) is defined by

$$V^a(t) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1, j \neq i}^n V_{ij}^a(t) \quad (10)$$

where $V_{ij}^a(t) \in \Re$ is the intervehicle collision avoidance potential function of UAV i and UAV j , given by

$$V_{ij}^a(t) = \begin{cases} \eta_j \epsilon_a^{(\rho_a - d_{ij}(t))/(\rho_a - R_a)}, & \text{if } j \in \mathcal{D}_i \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

with $\mathcal{D}_i$ and R_a as in (4), $\epsilon_a \in \Re$ a very small positive scalar, $0 < \rho_a < R_a$, and $\eta_j > 0$. It is noted that $V_{ij}^a(t) = 0$ when $d_{ij}(t) > R_a$, which indicates that the collision avoidance potential functions only work when the distance between two vehicles is less than or equal to the maximum collision detection distance R_a . Moreover, it is noted that $V_{ij}^a(t)$ increases exponentially as the distance $d_{ij}(t)$ decreases, and the potential function $V_{ij}^a(t)$ will tend to infinity if $d_{ij}(t)$ tends to zero (i.e., collision of UAVs i and j) when $\rho < R_a$ is selected to be arbitrarily close to R_a . Fig. 1 shows the relationship between $V_{ij}^a(t)$ and $d_{ij}(t)$.

The potential function $V^t(t)$ in (7) is defined by

$$V^t(t) = \sum_{i=1}^n \beta_i(t) V_i^t(t) \quad (12)$$

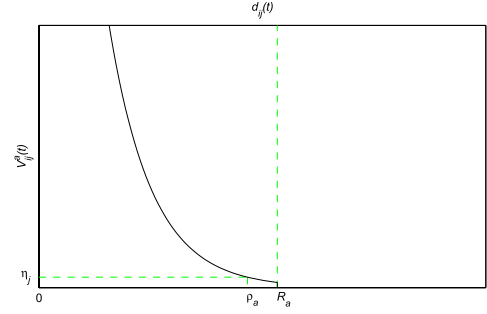


Fig. 1. Relationship between $V_{ij}^a(t)$ and $d_{ij}(t)$.

where $V_i^t(t) \in \Re$ is the potential function for vehicle i tracking the VL and is given by

$$V_i^t(t) = \frac{1}{2} [\mathbf{p}_i(t) - \mathbf{p}_v - (\mathbf{p}_{fi} - \mathbf{p}_{fv})]^T \times [\mathbf{p}_i(t) - \mathbf{p}_v - (\mathbf{p}_{fi} - \mathbf{p}_{fv})] \quad (13)$$

with $\mathbf{p}_v \in \Re^3$ the waypoint of the VL in NED frame and $\mathbf{p}_{fv} \in \Re^3$ the assigned position of the VL in the desired formation configuration. The weight $\beta_i(t) \geq 0$ satisfies

$$\beta_i(t) = \begin{cases} c_i > 0, & \text{if information from VL is received} \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

$$\sum_{i=0}^n \beta_i(t) > 0. \quad (15)$$

Remark 1: Note that (15) ensures that the VL's information is received by at least one UAV in the team. Then, with Assumption 1, the potential function $V(t) = 0$ ensures that the team of UAVs maintain the formation and track the VL with guaranteed intervehicle collision avoidance. As the potential function $V(t)$ is based on the relative positions and distances between neighbors, the derived control is distributed.

The following underlying assumption ensures that the collision avoidance set $\mathcal{D}_i$ ($i = 1, 2, \dots, n$) is empty when UAV i is in the neighborhood of its desired position as required by the formation.

Assumption 2: Intervehicle spacings of the desired formation $\mathbf{p}_f = [\mathbf{p}_{f1} \ \mathbf{p}_{f2}, \dots, \mathbf{p}_{fn}]^T$ are far larger than the maximum collision detection distance R_a .

Theorem 1: Consider a team of n VTOL UAVs satisfying Assumptions 1 and 2, and a predefined VL that is connected with at least one UAV in the team [i.e., (14) and (15) hold]. The team of UAVs asymptotically achieves the desired formation $\mathbf{p}_f = [\mathbf{p}_{f1} \ \mathbf{p}_{f2}, \dots, \mathbf{p}_{fn}]^T$ and follows the VL with guaranteed intervehicle collision avoidance if all the individual UAVs fly in the following corresponding speed:

$$\mathbf{v}_{ri}(t) = \mathbf{K}^f \left\{ \sum_{j \in \mathcal{N}_i} [\mathbf{p}_i(t) - \mathbf{p}_j(t) - (\mathbf{p}_{fi} - \mathbf{p}_{fj})] + \beta_i(t) [\mathbf{p}_i(t) - \mathbf{p}_v - (\mathbf{p}_{fi} - \mathbf{p}_{fv})] + \sum_{j \in \mathcal{D}_i} \eta_j \alpha_{ij}(t) \frac{\mathbf{p}_i(t) - \mathbf{p}_j(t)}{d_{ij}(t)} \right\} \quad i = 1, 2, \dots, n \quad (16)$$

where $\mathbf{p}_i(t) \in \Re^3$ is the position of the UAV i , $\mathbf{p}_v$ and $\mathbf{p}_{fv}$ are defined as in (13), $\mathbf{K}^f \in \Re^{3 \times 3}$ is a symmetric negative definite

matrix, $\mathcal{N}_i$ and $\mathcal{D}_i$ are defined in (1) and (4), respectively, $\eta_j > 0$, $d_{ij}(t)$ is as in (5), and

$$\alpha_{ij}(t) = -\frac{\epsilon_a^{(\rho_a - d_{ij}(t))/(\rho_a - R_a)} \ln \epsilon_a}{\rho_a - R_a} \quad (17)$$

with ϵ_a , ρ_a , and R_a defined as in (11).

Proof: Consider the potential function $V(t)$ described in (7)–(15), which can be rewritten as

$$V(t) = \sum_{i=1}^n \left\{ \beta_i(t) V_i^t(t) + \frac{1}{2} \sum_{j=1, j \neq i}^n [w_{ij}(t) V_{ij}^f(t) + V_{ij}^a(t)] \right\}.$$

Since $\beta_i(t)$ and $w_{ij}(t)$ are step functions, we have

$$\dot{V}(t) = \sum_{i=1}^n \Delta_i^T(t) \dot{\mathbf{p}}_i(t) \quad (18)$$

with

$$\begin{aligned} \Delta_i(t) &= \beta_i(t) \frac{\partial V_i^t(t)}{\partial \mathbf{p}_i(t)} + \sum_{j=1, j \neq i}^n w_{ij}(t) \frac{\partial V_{ij}^f(t)}{\partial \mathbf{p}_i(t)} \\ &\quad + \sum_{j=1, j \neq i}^n \frac{\partial V_{ij}^a(t)}{\partial \mathbf{p}_i(t)}. \end{aligned} \quad (19)$$

From (9), (11), and (13), we have

$$\frac{\partial V_{ij}^f(t)}{\partial \mathbf{p}_i(t)} = \mathbf{p}_i(t) - \mathbf{p}_j(t) - (\mathbf{p}_{fi} - \mathbf{p}_{fj}) \quad (20)$$

$$\frac{\partial V_{ij}^a(t)}{\partial \mathbf{p}_i(t)} = \begin{cases} \eta_j \alpha_{ij}(t) [\mathbf{p}_i(t) - \mathbf{p}_j(t)] / d_{ij}(t), & \text{if } j \in \mathcal{D}_i \\ 0, & \text{otherwise} \end{cases} \quad (21)$$

$$\frac{\partial V_i^t(t)}{\partial \mathbf{p}_i(t)} = \mathbf{p}_i(t) - \mathbf{p}_v - (\mathbf{p}_{fi} - \mathbf{p}_{fv}) \quad (22)$$

with $\alpha_{ij}(t)$ as in (17) and $d_{ij}(t)$ as in (5). Consequently

$$\sum_{j=1, j \neq i}^n w_{ij}(t) \frac{\partial V_{ij}^f(t)}{\partial \mathbf{p}_i(t)} = \sum_{j \in \mathcal{N}_i} [\mathbf{p}_i(t) - \mathbf{p}_j(t) - (\mathbf{p}_{fi} - \mathbf{p}_{fj})] \quad (23)$$

$$\sum_{j=1, j \neq i}^n \frac{\partial V_{ij}^a(t)}{\partial \mathbf{p}_i(t)} = \sum_{j \in \mathcal{D}_i} \eta_j \alpha_{ij}(t) \frac{\mathbf{p}_i(t) - \mathbf{p}_j(t)}{d_{ij}(t)}. \quad (24)$$

Substituting (22)–(24) into (16), we have

$$\mathbf{v}_{ri}(t) = \mathbf{K}^f \Delta_i(t), \quad i = 1, 2, \dots, n. \quad (25)$$

For $\dot{\mathbf{p}}_i(t) = \mathbf{v}_i(t) = \mathbf{v}_{ri}(t)$, substituting (25) into (18), we have

$$\dot{V}(t) = \sum_{i=1}^n \Delta_i^T(t) \mathbf{K}^f \Delta_i(t). \quad (26)$$

It should be highlighted that the introduction of $V_{ij}^a(t)$ is for the purpose of collision avoidance, which has higher priority than formation achievement and VL tracking. Hence, the parameters in $V_{ij}^a(t)$ are selected such that, upon achievement of formation, the collision avoidance sets $\mathcal{D}_i$ ($i = 1, 2, \dots, n$) are empty. Furthermore, when UAV i collides with UAV j , $V_{ij}^a(t)$ tends to be far larger than $V(0)$. Note that $V^a(0)$ can be small or zero as we can set the initial positions of UAVs.

Since $\mathbf{K}^f < 0$,¹ from (26), we have $\dot{V}(t) \leq 0$, hence, $V_{ij}^a(t) \leq V^a(t) \leq V(t) \leq V(0)$, which indicates that intervehicle collision can be avoided.

As $\dot{\mathbf{p}}_i(t) = \mathbf{v}_{ri}(t)$, (16) can be rewritten as

$$\dot{\mathbf{p}}(t) = \mathbf{K}[\mathbf{L}(t) + \mathbf{C}(t)][\mathbf{p}(t) - \mathbf{p}_f - \mathbf{1}_n \otimes (\mathbf{p}_v - \mathbf{p}_{fv})] + \Omega(t) \quad (27)$$

where

$$\mathbf{p}(t) \triangleq [\mathbf{p}_1^T(t) \mathbf{p}_2^T(t) \cdots \mathbf{p}_n^T(t)]^T \quad (28)$$

$$\mathbf{p}_f \triangleq [\mathbf{p}_{f1}^T \mathbf{p}_{f2}^T \cdots \mathbf{p}_{fn}^T]^T \quad (29)$$

$$\mathbf{C}^f(t) \triangleq \text{diag}\{\beta_1(t), \beta_2(t), \dots, \beta_n(t)\} \quad (30)$$

$$\mathbf{L}(t) \triangleq \mathbf{L}_G(t) \otimes \mathbf{I}_3 \quad (31)$$

$$\mathbf{K} \triangleq \mathbf{I}_n \otimes \mathbf{K}^f \quad (32)$$

$$\mathbf{C}(t) \triangleq \mathbf{C}^f(t) \otimes \mathbf{I}_3 \quad (33)$$

$$\Omega(t) \triangleq [\Omega_1^T(t) \Omega_2^T(t) \cdots \Omega_n^T(t)]^T \quad (34)$$

$$\Omega_i(t) \triangleq \mathbf{K}^f \sum_{j \in \mathcal{D}_i} \eta_j \alpha_{ij}(t) \frac{\mathbf{p}_i(t) - \mathbf{p}_j(t)}{d_{ij}(t)}. \quad (35)$$

It is noted that $\Omega_i(t)$ in (35) is the third term in (16) and $\Omega(t)$ is considered as the input to (27). Since $V_{ij}^a(t) \leq V(0)$, from (11), (17), and (35), we know that $\Omega_i(t)$ is bounded and so is $\Omega(t)$.

According to Assumption 1, the graph Laplacian $\mathbf{L}_G(t) \geq 0$ is with one zero eigenvalue and all its nonzero eigenvalues are positive. Namely, $\text{rank}(\mathbf{L}_G(t)) = n - 1$ and $\mathbf{1}_n^T \mathbf{L}_G(t) \mathbf{1}_n = 0$ with $\mathbf{1}_n = [1 \ 1 \cdots 1]^T$. Furthermore, from (30), we have $\mathbf{C}^f(t) \geq 0$ and $\mathbf{1}_n^T \mathbf{C}^f(t) \mathbf{1}_n > 0$ due to (14) and (15). Hence, $\mathbf{L}_G(t) + \mathbf{C}^f(t) > 0$. Using (31)–(33) with $\mathbf{K}^f < 0$, we have $\mathbf{K}(\mathbf{L}(t) + \mathbf{C}(t)) = (\mathbf{L}_G(t) + \mathbf{C}^f(t)) \otimes \mathbf{K}^f < 0$. Hence, the system (27) is asymptotically stable.

As (27) is asymptotically stable and $\Omega(t)$ is bounded, the spacing of any two UAVs will converge to a bounded neighborhood of the desired spacing as required by the desired formation. Since Assumption 2 ensures that the collision avoidance sets $\mathcal{D}_i$ ($i = 1, 2, \dots, n$) are empty in the neighborhood of the desired spacings, we conclude that the system (27) can drive $\mathcal{D}_i \rightarrow \emptyset$ for $i = 1, 2, \dots, n$, i.e., $\Omega(t) \rightarrow 0$. As a result, we have $\lim_{t \rightarrow \infty} \mathbf{p}(t) - \mathbf{p}_f = \mathbf{1}_n \otimes (\mathbf{p}_v - \mathbf{p}_{fv})$. Namely, the team of UAVs asymptotically achieves formation flight, tracks the VL, and avoids intervehicle collision. This completes the proof. $\diamond$

Remark 2: Theorem 1 presents a distributed control approach to generate the desired velocity for each UAV. If all UAVs fly in their respective desired velocities $\mathbf{v}_{ri}$ ($i = 1, 2, \dots, n$), it is guaranteed that the team of UAVs can perform formation flight/reconfiguration and track the given VL as well as avoid intervehicle collision.

Remark 3: It is noted that $\mathbf{p}_f$ defined in (29) describes the desired shape of formation with $\mathbf{p}_{fi} = \mathbf{T} \mathbf{p}_{fri}$, where $\mathbf{p}_{fr} = [\mathbf{p}_{fr1}^T \mathbf{p}_{fr2}^T \cdots \mathbf{p}_{frn}^T]^T$ is the reference shape of formation which describes a rigid body and $\mathbf{T}$ is the transformation matrix given by (36), shown at the top of the next page, with ψ_f , θ_f , and ϕ_f the Euler angles of the

¹For matrix M , $M > 0$ (< 0) represents that M is positive (negative) definite; $M \geq 0$ (≤ 0) represents that M is positive (negative) semidefinite.

$$\mathbf{T} = \begin{bmatrix} \cos \psi_f \cos \theta_f & \cos \psi_f \sin \theta_f \sin \phi_f - \sin \psi_f \cos \phi_f & \cos \psi_f \sin \theta_f \cos \phi_f + \sin \psi_f \sin \phi_f \\ \sin \psi_f \cos \theta_f & \sin \psi_f \sin \theta_f \sin \phi_f + \cos \psi_f \cos \phi_f & \sin \psi_f \sin \theta_f \cos \phi_f - \cos \psi_f \sin \phi_f \\ -\sin \theta_f & \cos \theta_f \sin \phi_f & \cos \theta_f \cos \phi_f \end{bmatrix} \quad (36)$$

reference formation shape as a rigid body in 3-D space. Similarly, the assigned position of the VL in the desired formation can be described by $\mathbf{p}_{fv} = \mathbf{T}\mathbf{p}_{fvr}$ with $\mathbf{p}_{fvr}$ the assigned position of VL in the reference formation shape. It is noted that the desired formation $\mathbf{p}_f$ is a step function of time. It varies only when the formation is required to vary from one shape to another or rotate from one pose to another in 3-D space (see Section III for details).

Remark 4: The larger the control gain $\mathbf{K}^f$ in (16) is, the more robust the system against the disturbance is. However, there is a tradeoff between the robustness and velocity constraints.

The above discusses the outer loop of the cascade control that generates the desired velocity for each UAV. In the following, the inner loop of the cascade control will be discussed to drive the velocity of each UAV to track its desired velocity $\mathbf{v}_{ri}(t)$ ($i = 1, 2, \dots, n$), based on a common velocity platform subject to measurement errors and dynamic constraints.

B. Robust Inner-Loop Control

Theorem 2: Under Assumption 1, all the velocities of the UAVs $\mathbf{v}_i(t) \in \mathbb{R}^3$ ($i = 1, 2, \dots, n$) converge to their corresponding desired velocities $\mathbf{v}_{ri}(t) \in \mathbb{R}^3$ ($i = 1, 2, \dots, n$) asymptotically if all the individual UAVs fly with the following corresponding acceleration:

$$\begin{aligned} \mathbf{u}_i(t) = & \mathbf{K}^v \sum_{j \in \mathcal{N}_i} [\mathbf{v}_i(t) - \mathbf{v}_{ri}(t) - \mathbf{v}_j(t) + \mathbf{v}_{rj}(t)] \\ & + \beta_i(t) \mathbf{K}^v [\mathbf{v}_i(t) - \mathbf{v}_{ri}(t)] + \dot{\mathbf{v}}_{ri}(t) \\ & i = 1, 2, \dots, n \end{aligned} \quad (37)$$

where $\mathbf{K}^v \in \mathbb{R}^{3 \times 3}$ is a symmetric negative definite matrix, $\beta_i(t) \in \mathbb{R}$ is as in (14) and (15), and $\mathcal{N}_i$ is as in (1).

Proof: Denote

$$\mathbf{v}(t) \triangleq \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \\ \vdots \\ \mathbf{v}_n(t) \end{bmatrix} \in \mathbb{R}^{3n}, \quad \mathbf{v}_r(t) \triangleq \begin{bmatrix} \mathbf{v}_{r1}(t) \\ \mathbf{v}_{r2}(t) \\ \vdots \\ \mathbf{v}_{rn}(t) \end{bmatrix} \in \mathbb{R}^{3n}. \quad (38)$$

Similar to the proof of Theorem 17, as $\dot{\mathbf{v}}_i = \mathbf{u}_i$, (37) can be rewritten as

$$\begin{aligned} \dot{\mathbf{v}}(t) = & \mathbf{K}_V \{\mathbf{L}(t)[\mathbf{v}(t) - \mathbf{v}_r(t)] + \mathbf{C}(t)[\mathbf{v}(t) - \mathbf{v}_r(t)]\} + \dot{\mathbf{v}}_r(t) \\ = & \mathbf{K}_V(\mathbf{L}(t) + \mathbf{C}(t))[\mathbf{v}(t) - \mathbf{v}_r(t)] + \dot{\mathbf{v}}_r(t) \end{aligned} \quad (39)$$

with $\mathbf{K}_V = \mathbf{I}_n \otimes \mathbf{K}^v$, $\mathbf{L}(t)$ and $\mathbf{C}(t)$ as in (31) and (33), respectively.

Since $\mathbf{K}_V(\mathbf{L}(t) + \mathbf{C}(t)) = (\mathbf{L}_{\mathcal{G}}(t) + \mathbf{C}^f(t)) \otimes \mathbf{K}^v < 0$, the system (39) is asymptotically stable and $\lim_{t \rightarrow \infty} \mathbf{v}(t) - \mathbf{v}_r(t) = 0$. Namely, the velocities of all individual UAVs asymptotically converge to their desired ones. This completes the proof. $\diamond$

Remark 5: It is noted that Theorem 2 proposes a consensus-based velocity tracking control approach for a team of UAVs. The first item in (37) achieves consensus of the velocity errors between the real UAV velocity $\mathbf{v}_i$ and its desired velocity $\mathbf{v}_{ri}$ ($i = 1, 2, \dots, n$) and the second item in (37) drives the consensus value on the velocity errors to zero.

After consideration of the control constraints and measurement errors, the closed inner loop system of the team of vehicles becomes

$$\mathbf{G} : \begin{cases} \dot{\mathbf{v}}(t) = \mathbf{K}_V(\mathbf{L}(t) + \mathbf{C}(t))[\mathbf{v}(t) - \mathbf{v}_r(t)] \\ \quad + \mathbf{L}_N \mathbf{w}_c(t) + \dot{\mathbf{v}}_r(t) \\ \mathbf{z}(t) = \mathbf{v}(t) - \mathbf{v}_r(t) \end{cases} \quad (40)$$

where $\mathbf{w}_c(t) \in \mathbb{R}^{3n}$ is the bounded disturbance due to control constraints and measurement errors, and $\mathbf{z}(t)$ is the performance output. Since uniform disturbances do not affect formation flying and intervehicle collision avoidance of a team of UAVs (even though they do affect the performance of the team of UAVs tracking the VL), here $\mathbf{L}_N = \mathbf{L}_n \otimes \mathbf{I}_3$ with $\mathbf{L}_n = n\mathbf{I}_n - \mathbf{1}_n \mathbf{1}_n^T$ is introduced to remove uniform disturbances and only consider nonuniform disturbances.

In the following theorem, an H_∞ control design approach is proposed to determine the gain matrix $\mathbf{K}^v$ in (37).

Theorem 3: Assume that Assumption 1 holds. Consider the system $\mathbf{G}$ in (40). For a given scalar $\mu > 0$, suppose that there exist a matrix $\mathbf{Y} \in \mathbb{R}^{3 \times 3}$ and a symmetric positive definite matrix $\mathbf{Q} \in \mathbb{R}^{3 \times 3} > 0$ satisfying

$$\begin{bmatrix} (\mathbf{L}_{\mathcal{G}}(t) + \mathbf{C}^f(t)) \otimes (\mathbf{Y} + \mathbf{Y}^T) & \mathbf{L}_n \otimes \mathbf{Q} & \mathbf{I}_{3n} \\ \mathbf{L}_n \otimes \mathbf{Q} & -\mu \mathbf{I}_{3n} & 0 \\ \mathbf{I}_{3n} & 0 & -\mathbf{I}_{3n} \end{bmatrix} < 0 \quad (41)$$

$\forall \mathbf{L}_{\mathcal{G}}(t) \in \mathcal{L} \quad \forall \mathbf{C}^f(t) \in \mathcal{C}$

where $\mathcal{L}$ is the set of Laplacian matrices for all possible connected communication network $\mathcal{G}$ and $\mathcal{C}$ is the set of all possible matrices $\mathbf{C}^f(t)$ defined in (30). Then the system (40) is asymptotically stabilized by the controller

$$\mathbf{K}^v = \mathbf{Q}^{-1} \mathbf{Y} \quad (42)$$

with $\|\mathbf{G}\|_\infty < \mu$.

Proof: The proof is quite direct and is omitted here. $\diamond$

Remark 6: Theorem 3 gives a way to find robust inner-loop control laws against control constraints and measurement errors.

C. Heading Control

For a quadrotor UAV, its yaw angle is independent of its flight direction. To make the yaw angles of the team of UAVs converge to the yaw angle ψ_v of the VL, the reference yaw rate of the UAV i is given by

$$\dot{\psi}_{ri}(t) = \mathbf{K}^\psi \left\{ \sum_{j \in \mathcal{N}_i} [\psi_i(t) - \psi_j(t)] + \beta_i(t) [\psi_i(t) - \psi_v] \right\} \quad i = 1, 2, \dots, n \quad (43)$$

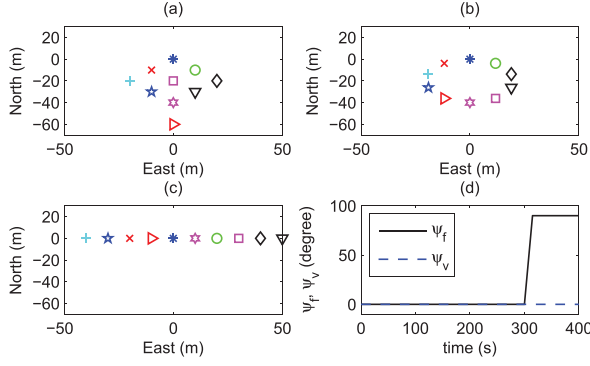


Fig. 2. Reference formation shape in simulation. (a) $t < 100$ s. (b) $100 \leq t < 200$ s. (c) $t \geq 200$ s. (d) Yaw angle.

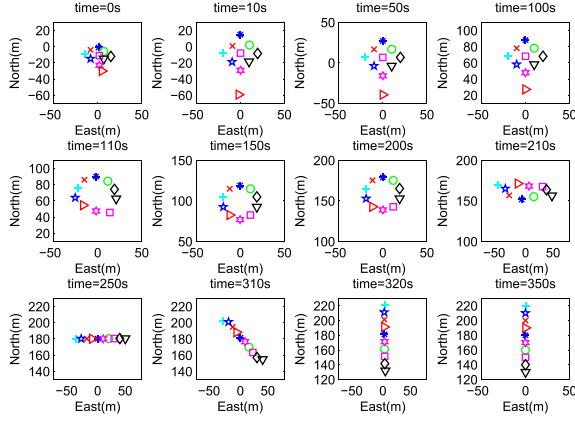


Fig. 3. Simulation results of X-Y view of formation.

where $\mathbf{K}^\psi \in \mathbb{R}^{3 \times 3} < 0$ is a symmetric negative definite matrix and ψ_i is the yaw angle of UAV i .

III. SIMULATION OF FORMATION CONTROL

In this simulation, ten quadrotor UAVs are used to demonstrate formation flight and reconfiguration. The controller parameters used in the simulations are $\mathbf{K}^f = -0.1\mathbf{I}_3$, $\mathbf{K}^v = -0.35\mathbf{I}_3$, $\mathbf{K}^\psi = -0.5$, $\epsilon_a = 0.5$, $\rho_a = 7.8$ m, $R_a = 8$ m, $c_i = 1$, and $\eta_j = 1$. It is set in the simulation that one UAV cannot receive the information from another UAV if the distance between them is larger than 40 m.

The reference formation shapes $\mathbf{p}_{fr}$ for different time periods are shown in Fig. 2(a)–(c), where the points represent the positions of UAVs in the reference formation shape. The assigned position $\mathbf{p}_{fvr}$ of VL in the reference formation shape is at (0, 0, -5), which coincides with the * UAV in Fig. 2(a)–(c). The Euler angle of the reference formation shape in 3-D space is given by $\theta_f = \phi_f = 0$ and ψ_f , as shown in Fig. 2(d). In this simulation, the VL flies along a straight line from (0, 0, -5) to (180, 0, -5) in NED frame with the yaw angle $\psi_v = 0$ as shown in Fig. 2(d).

Figs. 3 and 4 show the simulation results of the ten quadrotor UAVs controlled by the proposed formation control approach. From Fig. 3, it is observed that the team of ten UAVs can form/reconfigure the desired formations and maintain them in motion. As the position of the VL in the reference formation shape is the same as that of the * UAV, it is observed that the * UAV always follows the path of the VL from

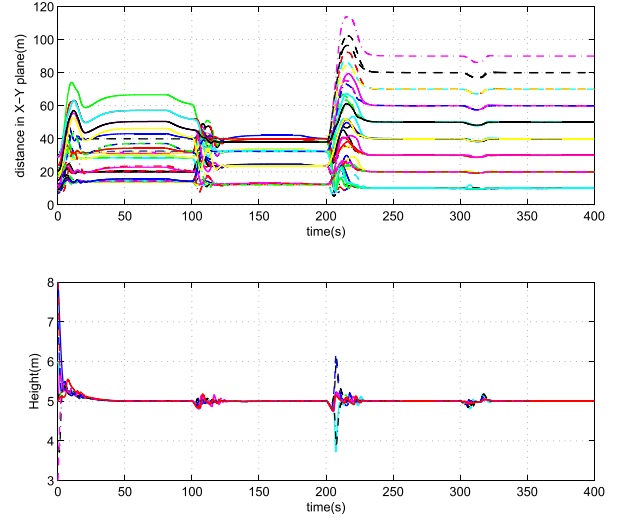


Fig. 4. Simulation results of intervehicle distances and UAV heights.



Fig. 5. Quadrotor UAV.

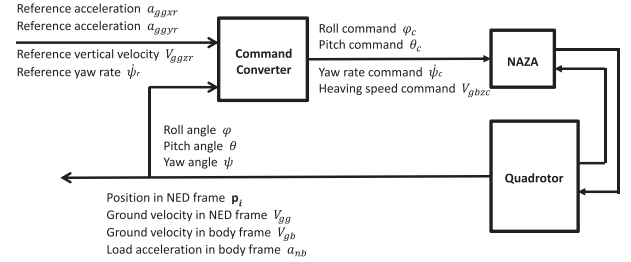


Fig. 6. Extended quadrotor UAV system.

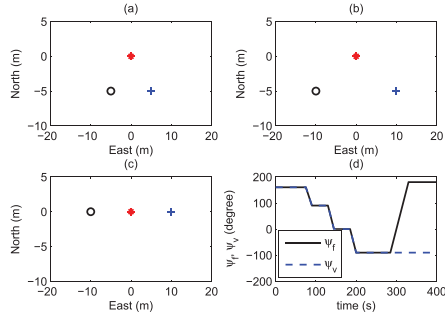
(0, 0, -5) to (180, 0, -5) in NED frame very well, except during the transient reconfiguration period around $t = 210$ s. Fig. 4 shows that the intervehicle distances between any two UAVs are all larger than 5 m. Hence, there is no intervehicle collision among these UAVs. The video of this simulation can be found at <http://youtu.be/JIUrOm9MZwQ>.

IV. FLIGHT TEST OF FORMATION CONTROL

The UAVs used in the flight tests are the commercial quadrotors shown in Fig. 5. Each UAV is equipped with NAZA that implements the attitude and altitude control. NAZA is connected with the formation controller via a command converter shown in Fig. 6, where the reference acceleration $[a_{ggrx} \ a_{ggyr} \ a_{ggzr}]^T = \mathbf{u}_i(t)$ is given by (37) and the reference velocity $[V_{ggrx} \ V_{ggyr} \ V_{ggzr}]^T = \mathbf{v}_{ri}(t)$ is given by (16) [16].



Fig. 7. Formation flight test of three quadrotor UAVs.

Fig. 8. Reference formation shape in flight test. (a) $t < 245$ s. (b) $245 \text{ s} \leq t < 265$ s. (c) $t \geq 265$ s. (d) Yaw angle.

The dynamic constraints of the quadrotor UAVs are as follows: $|\phi| \leq 0.3$ rad, $|\theta| \leq 0.3$ rad, $|\dot{\phi}| \leq 0.9$ rad/s, $|\dot{\theta}| \leq 0.9$ rad/s, $|\dot{\psi}| \leq 1.8$ rad/s, and $0 \leq V \leq 5$ m/s, where ϕ , θ , ψ , and V are the roll, pitch, and yaw angles and the cruise speed of the UAVs, respectively.

In the flight test of three quadrotor UAVs, the communication rate is 10 Hz. The formation control loop including outer loop and inner loop control runs at 10 Hz, while the stability control loop [16] runs at 50 Hz. Fig. 7 shows a photo of the three-UAV formation test. The controller parameters used in the test are 1) $\mathbf{K}^f = -0.1\mathbf{I}_3$, 2) $\mathbf{K}^v = -0.35\mathbf{I}_3$, 3) $\mathbf{K}^w = -0.5$, 4) $\epsilon_a = 0.1$, 5) $\rho_a = 4.5$ m, 6) $R_a = 5$ m, and 7) $c_i = \eta_j = 1$.

The procedure for the formation test is as follows. First, the ground station gives a takeoff command to the three UAVs. The three UAVs take off automatically and hover at the specified positions. Then at $t = 0$, the ground station gives a formation command to the three UAVs. The three UAVs start to complete the whole formation test process: they first form the triangle formation with its reference formation shape as shown in Fig. 8(a) and make translational and rotational movements with $\theta_f = \phi_f = 0$ and $\psi_f = 160^\circ \rightarrow -90^\circ$ as shown in Fig. 8(d). Then at $t = 245$ s, the triangle formation enlarges to a big one. Its reference formation shape is shown in Fig. 8(b) with $\theta_f = \phi_f = 0$ and $\psi_f = -90^\circ$. At $t = 265$ s, the big triangle formation is reconfigured to the alongside formation. Its reference formation shape is shown in Fig. 8(c) with $\theta_f = \phi_f = 0$ and $\psi_f = -90^\circ$. Then at $t = 285$ s, this alongside formation rotates 270° clockwise around UAV 2 (* UAV) to form the tandem formation. Finally, at $t = 335$ s, the tandem formation moves westward.

The position of the VL in the reference formation shape is (0, 0, -5), which coincides with that of * UAV (i.e., UAV 2) in Fig. 8(a)–(c). The path of the VL is shown with a dashed-dotted line in Fig. 10, and its heading angle ψ_v in Fig. 8(d).

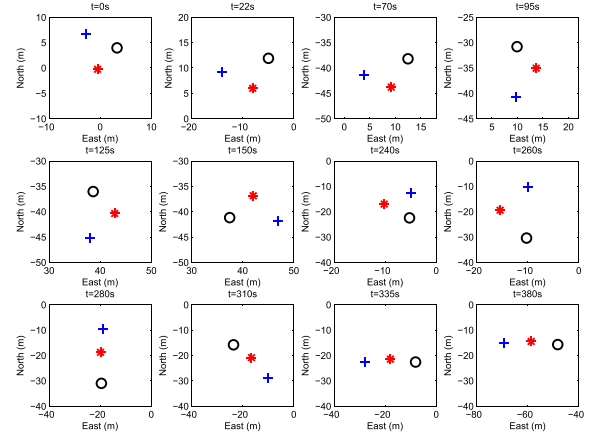


Fig. 9. X-Y view of formation in flight test.

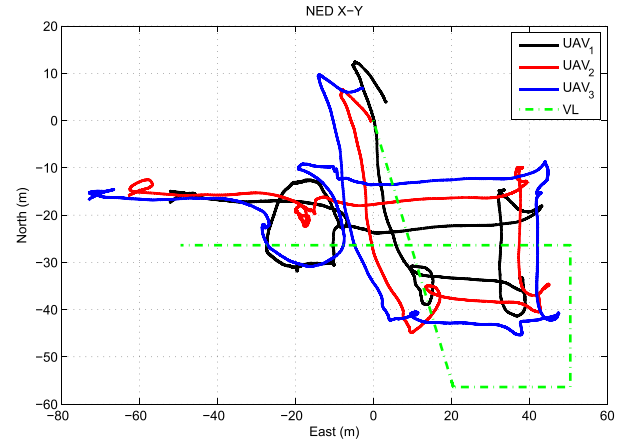


Fig. 10. X-Y view of trajectories in flight test.

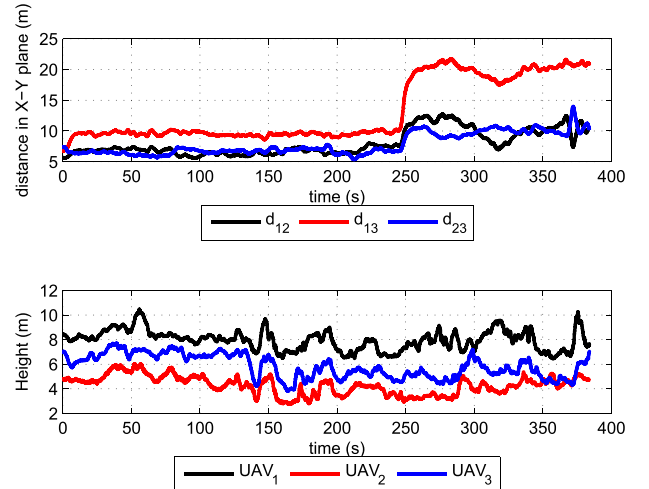


Fig. 11. Distances and heights of UAVs in flight test.

The results of the formation flight test are shown in Figs. 9–12. From Fig. 9, it is observed that the three UAVs can form/reconfigure the desired formations and maintain them quite well even when they move. Fig. 10 shows the horizontal trajectories of the three UAVs together with that of the VL. It is noted that the trajectory of UAV 2 drifts from the path of the VL due to the strong northwest wind. However, the formation shape can still be maintained well. Fig. 11 shows that the distances between any two UAVs are larger than 5 m

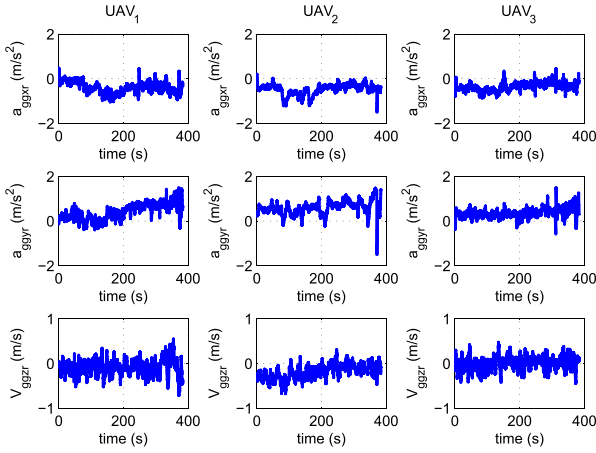


Fig. 12. Control inputs of extended quadrotor UAVs in flight test.

and the maximum steady-state horizontal distance error is around 2.5 m and the maximum steady-state height error is around 3 m. It should be mentioned that the positioning accuracy of the low-cost GPS is more than 1 m for the horizontal position and 3 m for height. The control inputs of the extended quadrotors are shown in Fig. 12, where no saturation is observed. The video of this flight test can be found at <http://youtu.be/eB1VFx1vwBM>.

The formation flight test validates the overall design of the formation control system. The test results show that the formation shape can be robustly maintained at reasonable deviation despite the low accuracy of GPS, low accuracy of the linear mathematic model of the UAVs, and strong wind disturbance. To improve the performance of tracking the VL, one way is to increase the weight c_i in (14) and another way is to introduce integral control of the tracking error.

V. CONCLUSION

In this brief, a unified distributed robust formation and reconfiguration control approach is proposed for VTOL UAVs with dynamic constraints. Its effectiveness is demonstrated by simulation and flight test. The proposed approach is based on undirected switching communication network. In the future, it will be extended to directed switching network (i.e., cameras are used to estimate the relative position) and fixed-wing UAVs.

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